

DBMS Practical-2

Question1 Create Table

```
mysql> create table student_marks1 (  
-> roll_no int primary key,  
-> name varchar(50),  
-> subject varchar(50),  
-> marks decimal(5,2)  
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> show tables;
```

```
+-----+  
| Tables_in_task2 |  
+-----+  
| student_marks1 |  
+-----+
```

1 row in set (0.00 sec)

```
mysql> desc student_marks1;
```

Field	Type	Null	Key	Default	Extra
roll_no	int	NO	PRI	NULL	
name	varchar(50)	YES		NULL	
subject	varchar(50)	YES		NULL	
marks	decimal(5,2)	YES		NULL	

4 rows in set (0.00 sec)

Question2. Insert Data

```
mysql> insert into student_marks1 (roll_no, name, subject, marks) values  
-> (1, 'ravi', 'math', 85.50),  
-> (2, 'sita', 'math', 92.75),  
-> (3, 'anil', 'math', 78.40),  
-> (4, 'priya', 'math', 88.90),  
-> (5, 'vijay', 'math', 80.25),  
-> (6, 'subbusir', 'aws', 98.50),  
-> (7, 'dwaraka', 'dbms', 95.75),  
-> (8, 'ranjani', 'english', 97.40),  
-> (9, 'kavithamam', 'aws1', 99.90),  
-> (10, 'seetha', 'azure', 82.25);
```

Query OK, 10 rows affected (0.01 sec)

Records: 10 Duplicates: 0 Warnings: 0

Question3. Simple Select

```
mysql> select * from student_marks1;
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
3	anil	math	78.40
4	priya	math	88.90
5	vijay	math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.00 sec)

mysql>
```

Question4. Aggregate Functions

```
mysql> select count(*) as total_students from student_marks1;
```

total_students
10

```
1 row in set (0.01 sec)

mysql> select sum(marks) from student_marks1;
```

sum(marks)
899.60

```
1 row in set (0.01 sec)

mysql> select avg(marks) from student_marks1;
```

avg(marks)
89.960000

```
1 row in set (0.00 sec)

mysql> select max(marks) from student_marks1;
```

max(marks)
99.90

```
1 row in set (0.01 sec)

mysql> select min(marks) from student_marks1;
```

min(marks)
78.40

```
1 row in set (0.00 sec)

mysql> |
```

Question5. Filtering with WHERE

```
mysql> select * from student_marks1
-> where marks > 85;
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
4	priya	math	88.90
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

7 rows in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks >= 90;
```

roll_no	name	subject	marks
2	sita	math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

5 rows in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks < 80;
```

roll_no	name	subject	marks
3	anil	math	78.40

1 row in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks between 80 and 90;
```

roll_no	name	subject	marks
1	ravi	math	85.50
4	priya	math	88.90
5	vijay	math	80.25
10	seetha	azure	82.25

4 rows in set (0.01 sec)

```
mysql> select * from student_marks1
-> where name like 'p%';
```

roll_no	name	subject	marks
4	priya	math	88.90

1 row in set (0.00 sec)

```
mysql> select * from student_marks1
-> where name in ('ravi', 'sita', 'vijay');
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
5	vijay	math	80.25

3 rows in set (0.00 sec)

```
mysql> select * from student_marks1
-> where marks > 85
-> and (subject = 'math' or name like 'p%');
```

roll_no	name	subject	marks
1	ravi	math	85.50
2	sita	math	92.75
4	priya	math	88.90

3 rows in set (0.00 sec)

```
mysql> |
```

Question6. Update Queries

```
mysql> update student_marks1
-> set marks = 90.00
-> where roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from student_marks1
-> where roll_no = 3;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      3 | anil | math    | 90.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql> update student_marks1
-> set subject = 'remedial math'
-> where marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
mysql> select * from student_marks1
-> where marks < 80;
Empty set (0.00 sec)
```

```
mysql> |
```

Question7. Delete Queries

```
mysql> delete from student_marks1
-> where roll_no = 5;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> select * from student_marks1;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
|      1 | ravi   | math    | 85.50 |
|      2 | sita   | math    | 92.75 |
|      3 | anil   | math    | 90.00 |
|      4 | priya  | math    | 88.90 |
|      6 | subbusir | aws    | 98.50 |
|      7 | dwaraka | dbms    | 95.75 |
|      8 | ranjani | english | 97.40 |
|      9 | kavithamam | aws1 | 99.90 |
|     10 | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

```
mysql> delete from student_marks1
-> where marks < 75;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> select * from student_marks1;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
|      1 | ravi   | math    | 85.50 |
|      2 | sita   | math    | 92.75 |
|      3 | anil   | math    | 90.00 |
|      4 | priya  | math    | 88.90 |
|      6 | subbusir | aws    | 98.50 |
|      7 | dwaraka | dbms    | 95.75 |
|      8 | ranjani | english | 97.40 |
|      9 | kavithamam | aws1 | 99.90 |
|     10 | seetha | azure   | 82.25 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

Question8. Sorting

```
mysql> select * from student_marks1  
-> order by marks asc;
```

roll_no	name	subject	marks
10	seetha	azure	82.25
1	ravi	math	85.50
4	priya	math	88.90
3	anil	math	90.00
2	sita	math	92.75
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
6	subbusir	aws	98.50
9	kavithamam	aws1	99.90

9 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> order by marks desc;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	sita	math	92.75
3	anil	math	90.00
4	priya	math	88.90
1	ravi	math	85.50
10	seetha	azure	82.25

```
mysql> select * from student_marks1  
-> order by name asc;
```

roll_no	name	subject	marks
3	anil	math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	priya	math	88.90
8	ranjani	english	97.40
1	ravi	math	85.50
10	seetha	azure	82.25
2	sita	math	92.75
6	subbusir	aws	98.50

9 rows in set (0.00 sec)

```
mysql> select * from student_marks1  
-> order by name desc;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	sita	math	92.75
10	seetha	azure	82.25
1	ravi	math	85.50
8	ranjani	english	97.40
4	priya	math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	anil	math	90.00

9 rows in set (0.00 sec)

```
mysql> |
```

Question9. Group By

```
mysql> select subject, sum(marks) as total_marks
-> from student_marks1
-> group by subject;
+-----+-----+
| subject | total_marks |
+-----+-----+
| math   | 357.15      |
| aws    | 98.50       |
| dbms   | 95.75       |
| english| 97.40       |
| aws1   | 99.90       |
| azure  | 82.25       |
+-----+-----+
6 rows in set (0.02 sec)

mysql> select subject, avg(marks) as avg_marks
-> from student_marks1
-> group by subject;
+-----+-----+
| subject | avg_marks |
+-----+-----+
| math   | 89.287500 |
| aws    | 98.500000 |
| dbms   | 95.750000 |
| english| 97.400000 |
| aws1   | 99.900000 |
| azure  | 82.250000 |
+-----+-----+
6 rows in set (0.00 sec)

mysql> select subject, count(*) as num_students
-> from student_marks1
-> group by subject;
+-----+-----+
| subject | num_students |
+-----+-----+
| math   | 4            |
| aws    | 1            |
| dbms   | 1            |
| english| 1            |
| aws1   | 1            |
| azure  | 1            |
+-----+-----+
6 rows in set (0.00 sec)

mysql> |
```

Question10. Having

```
mysql> select subject, avg(marks) as avg_marks
-> from student_marks1
-> group by subject
-> having avg(marks) > 90;
+-----+-----+
| subject | avg_marks |
+-----+-----+
| aws    | 98.500000 |
| dbms   | 95.750000 |
| english| 97.400000 |
| aws1   | 99.900000 |
+-----+-----+
4 rows in set (0.00 sec)

mysql> select subject, count(*) as num_students
-> from student_marks1
-> group by subject
-> having count(*) > 1;
+-----+-----+
| subject | num_students |
+-----+-----+
| math   | 4            |
+-----+-----+
1 row in set (0.00 sec)

mysql> |
```